Gilman's Alternating Quotients, Ground Down

Delta words $\to$ intersecting fibers $\to$ a 3-cycle

What has disappeared. The usual proof first establishes double transitivity and then invokes a minimal-degree theorem of Bochert. Neither is needed. A delta word gives a permutation supported on one coordinate fiber. Two such fibers meet in exactly one point, so the commutator of the two permutations is a literal 3-cycle. The normal closure of these 3-cycles is visibly transitive, and a transitive group generated by 3-cycles is the full alternating group.

1 The only word lemma

Let F_s be free of finite rank, let G be a finite nonabelian simple group, and let

$$\rho : F_s \twoheadrightarrow G, \qquad K = \ker \rho.$$

Lemma 1 (Delta word)**.** *For every $c \in G$ there is a word $W \in F_s$ such that*

$$\rho(W) = c, \qquad \mu(W) = 1$$

for every homomorphism $\mu : F_s \to G$ with $\ker \mu \neq K$.

Proof. List the distinct kernels of homomorphisms $F_s \to G$, other than K, as $L_1, \ldots, L_t$. First observe that

$$\rho(L_i) = G \qquad (1 \leq i \leq t).$$

Indeed, $L_i \not\subseteq K$: otherwise F_s/L_i, which is isomorphic to a subgroup of G, would surject onto $F_s/K \cong G$; finiteness would force $L_i = K$. Thus $\rho(L_i)$ is a nontrivial normal subgroup of the simple group G.

Put $M_0 = F_s$ and $M_i = M_{i-1} \cap L_i$. Inductively $\rho(M_{i-1}) = G$, and, since M_{i-1} and L_i are normal,

$$[M_{i-1}, L_i] \leq M_i.$$

But G is perfect, so

$$\rho([M_{i-1}, L_i]) = [\rho(M_{i-1}), \rho(L_i)] = [G, G] = G.$$

Hence $\rho(M_i) = G$ for every i. Choose $W \in M_t$ with $\rho(W) = c$. □

This commutator induction is the whole algebraic content of Gilman's separating-word lemma.

2 Two intersecting fibers make a 3-cycle

Let

$$F_5 = \langle x_1, \ldots, x_5 \rangle, \qquad \Gamma = \mathrm{SAut}(F_5).$$

We use the right and left Nielsen transvections

$$R_{ij} : x_i \mapsto x_i x_j, \qquad L_{ij} : x_i \mapsto x_j x_i \qquad (i \neq j),$$

with the other generators fixed. These transvections and their inverses generate Γ.

Fix a finite 2-generated nonabelian simple group

$$G = \langle a, b \rangle.$$

Let $X(G)$ be the set of generating 5-tuples of elements of G, modulo applying one common automorphism of G to all five entries. Nielsen transformations act on $X(G)$. Let

$$o = [a, b, 1, 1, 1], \qquad \Omega = \Gamma o, \qquad N = |\Omega|,$$

and let $H \leq \mathrm{Sym}(\Omega)$ be the image of Γ.

Define two coordinate fibers

$$B_5 = \{[a, b, 1, 1, g] : g \in G\}, \qquad B_4 = \{[a, b, 1, g, 1] : g \in G\}.$$

Both lie in Ω: write g as a word in a, b and build it in the indicated coordinate using transvections. Their displayed points are distinct, since an automorphism fixing a and b is the identity. Thus

$$|B_5| = |G|, \qquad B_4 \cap B_5 = \{o\}, \qquad N \geq |G|.$$

Apply Lemma 1 to the epimorphism

$$F_4 \twoheadrightarrow G, \qquad (x_1, x_2, x_3, x_4) \mapsto (a, b, 1, 1).$$

For $c \in G$, let W_c be the resulting word and define

$$\sigma_c : \quad x_5 \mapsto x_5 W_c(x_1, x_2, x_3, x_4).$$

This is a product of right transvections, hence belongs to Γ. On Ω it is supported exactly on B_5, where it acts by

$$[a, b, 1, 1, g] \mapsto [a, b, 1, 1, gc].$$

Indeed, outside B_5 the first four entries define a homomorphism with a different kernel, so W_c evaluates to 1; equality of kernels, conversely, lets us normalize the first four entries to $(a, b, 1, 1)$ by a common automorphism of G.

Similarly, using the variables x_1, x_2, x_3, x_5, we obtain $\tau_d \in \Gamma$ supported exactly on B_4, acting there by right multiplication by d.

We use the following one-line permutation fact.

Lemma 2 (Singleton-intersection commutator)**.** *If permutations α, β have supports meeting in exactly one point p, then*

$$[\alpha, \beta] = \alpha^{-1}\beta^{-1}\alpha\beta$$

is the 3*-cycle*

$$\big(p, \alpha^{-1}p, \beta^{-1}p\big).$$

Proof. The three displayed points cycle in the indicated order; every other point lies outside one support, and its image under the other never hits the intersection point, so it is fixed by the commutator. □

Fix $d \neq 1$. For every $g \neq 1$, Lemma 2 shows that

$$\kappa_g = [\sigma_{g^{-1}}, \tau_d]$$

is a 3-cycle whose support contains both o and $[a, b, 1, 1, g]$.

3 The normal closure is automatically alternating

Let $L \triangleleft H$ be the subgroup generated by all H-conjugates of the cycles κ_g $(g \neq 1)$. Then

$$B_5 \subseteq Lo.$$

Now consider the fixed "star" set of Nielsen generators

$$\mathcal{S} = \{R_{i5}^{\pm 1}, R_{5i}^{\pm 1}, L_{i5}^{\pm 1}, L_{5i}^{\pm 1} : 1 \leq i \leq 4\}.$$

It generates Γ. Indeed, for distinct $i, j < 5$,

$$R_{ij} = R_{5j}^{-1} R_{i5}^{-1} R_{5j} R_{i5}, \qquad L_{ij} = L_{5j}^{-1} L_{i5}^{-1} L_{5j} L_{i5}.$$

Every $s \in \mathcal{S}$ sends o into B_5: a transvection using x_5 as multiplier fixes o, while one changing x_5 changes only the fifth entry.

Consequently the L-orbit Lo is $\mathcal{S}$-invariant. Indeed, if $x = \ell o$ with $\ell \in L$ and $s \in \mathcal{S}$, then

$$sx = (s\ell s^{-1})(so) \in Lo,$$

because $L \triangleleft H$ and $so \in B_5 \subseteq Lo$. Since $\mathcal{S}$ generates H and $\Omega = Ho$, we get

$$Lo = \Omega.$$

Thus L is transitive and is generated by 3-cycles.

Lemma 3. *A transitive permutation group generated by 3-cycles is the full alternating group.*

Proof. Make a 3-uniform hypergraph whose edges are the supports of the generating cycles. Its connected components are exactly the group orbits, so the hypergraph is connected. Add its edges one at a time, each new edge meeting the vertices already reached. Inductively the cycles already added generate the alternating group on the old vertices. If a new edge adds one vertex, its cycle together with the old alternating group generates the alternating group after adjoining that vertex. If it adds two vertices, conjugate its cycle so that its one old vertex is replaced by a different old vertex; multiplying the two cycles (one inverted) gives a 3-cycle containing two old vertices and one new one. Add that new vertex first, and then the other. Hence all of A_N is generated. Since every generator is even, the group is exactly A_N. □

Lemma 3 gives $L = A_N$, hence $A_N \leq H \leq S_N$. In fact $H = A_N$: every Nielsen transvection is itself a commutator (use a third index in the displayed star identity), so Γ is perfect; therefore its quotient H is perfect and cannot be S_N. We have proved:

Theorem 4 (Gilman, in the form needed here)**.** *For every finite 2-generated nonabelian simple group G, there is an $N \geq |G|$ and a surjection*

$$\mathrm{SAut}(F_5) \twoheadrightarrow A_N.$$

4 The expander corollary

Corollary 5. *There is an infinite sequence $N_1 < N_2 < \cdots$ for which the groups A_{N_i} have bounded-size generating sets with a uniform spectral gap.*

Proof. For odd $m \geq 5$, take $G = A_m$. It is nonabelian simple and is generated by

$$(1\ 2\ \cdots\ m) \quad \text{and} \quad (1\ 2\ 3);$$

the conjugates of the 3-cycle by the long cycle have connected supports and hence generate A_m by Lemma 3. Theorem 4 gives

$$\pi_m : \mathrm{SAut}(F_5) \twoheadrightarrow A_{N_m}, \qquad N_m \geq |A_m| \longrightarrow \infty.$$

Kaluba–Nowak–Ozawa proved that $\mathrm{SAut}(F_5)$ has property (T). Hence the images under π_m of any one fixed finite generating set of $\mathrm{SAut}(F_5)$ have a uniform Kazhdan constant, and therefore a uniform Cayley-graph spectral gap. Pass to a subsequence with strictly increasing N_m. □

Summary. Apart from property (T) in the final corollary, the proof consists only of: normal subgroups of a simple group; the inclusion $[M, L] \leq M \cap L$ for normal subgroups; elementary Nielsen identities; and the fact that connectedly overlapping 3-cycles generate an alternating group. Double transitivity and Bochert are gone.

References

[1] R. Gilman, *Finite quotients of the automorphism group of a free group*, Canad. J. Math. **29** (1977), 541–551.

[2] M. Kaluba, P. W. Nowak, and N. Ozawa, $\mathrm{Aut}(F_5)$ *has property* (T), Math. Ann. **375** (2019), 1169–1191.